

Subject Description Form

Subject Code	LSGI4104
Subject Title	Geomatics Project Management
Credit Value	2
Level	4
Pre-requisites	LSGI2374 Surveying; and LSGI2375 Field Scheme (Surveying); and LSGI2223 Geographic Information Science
Objectives	This subject aims at the project management issues of engineering surveying, land boundary surveying, utility surveying and geographic information systems. The objectives of the course will be to familiarize the students with both the scope and functions of managerial operations associated with cost estimation, tendering, contract administration, data quality management, building information modelling information management and policy, formal report writing and so on. The teaching and learning adopted will help students develop critical and creative thinking
Intended Learning Outcomes	At the end of this subject students who gain a pass will be able to: 1. Understand the contemporary environment for geomatics business in Hong Kong (L2) 2. Grasp the essential procedures of conducting a geomatics project or business (L3) 3. Produce proper documentation of a geomatics project (L4) 4. Communicate and compromise with other professionals (e.g. government departments, contractor and real estate developer) in a construction project life-cycle (L4) 5. Apply principles of project accounting to manage budgeting, costing, and financial control in geomatics projects. (L3)
Subject Synopsis/ Indicative Syllabus	A. Geomatics Projects / Business • Types of geomatics project – engineering survey, land boundary survey, utility management, Geo-IT • Operations management B. Management of Geomatics Projects • Tendering process • Theme or objective identification • Project development lifecycle • Project management methodology

	development, language improvement, and the use of Gen-AIs must be referenced/acknowledged in their submitted materials) • A group project working on a particular /assigned aspect of geomatics is to let students gain experience on the essential procedures and proper documentary writings. (Students should be noticed that Gen-AIs can be used only for knowledge searching, idea development, language improvement, and the use of Gen-AIs must be referenced/acknowledged in their submitted materials) • An end-of-semester examination is to test students' knowledge learnt from the whole subject and to test their written/expression skills.	
Student Study Effort Expected	Class contact:	
	▪ Lecture/tutorial	26 Hrs.
	Other student study effort:	
	▪ literature review and web surfing	10 Hrs.
	▪ mini-project	20 Hrs.
	▪ self-study and revision	20 Hrs.
	Total student study effort	76 Hrs.
Reading List and References	1. Zimmerer, T. and Scarborough, N. (2005), <i>Essentials of Entrepreneurship and Small Business Management</i>, 4th ed., Pearson Education. 2. Course, L., Hubbard, D. and Wong, E. (2003), <i>Butterworths Hong Kong Contract Law Handbook</i>, Hong Kong: Lexis Nexis Butterworths. 3. Willoughby, P. and Wilkinson, M. (1995), <i>Registration of Titles in Hong Kong</i>, Hong Kong: Butterworth. 4. SMO (2005), <i>District Survey Office Technical Manual</i>. Hong Kong: Survey and Mapping Office, Lands Department. 5. Robillard, R., Wilson, D. and Brown, C. (2002), <i>Evidence and Procedures for Boundary Location</i>, 4th ed., New York: Wiley. 6. Harmon, J. (2003), <i>The Design and Implementation of Geographic Information Systems</i>, John Wiley & Sons. 7. Aronoff, S. (1989), <i>Geographic Information Systems: A Management Perspective</i>, WDL Publications. 8. Huxhold, W. (1994), <i>Managing Geographic Information System Projects</i>, New York: Oxford University Press.	

	9. HKEDC (1996, http://www.icac.org.hk/hkedc), Ethics for Professionals (Architecture, Engineering & Surveying), Hong Kong Ethics Development Centre, ICAC, HKSAR. 10. OCGIO (2012) <i>User Guide on PRINCE</i>, Office of the Government Chief Information Officer, HKSAR 11. OCGIO (2012) <i>An Introduction to Structured Systems Analysis & Design Methodology (SSADM)</i>, Office of the Government Chief Information Officer, HKSAR 12. Savory S (2010) <i>Journal of Building Information Modeling</i>, The National Institute of Building Sciences. Washington, DC. 13. Ruben de Laat and Leon Van Berlo (2011) Integration of BIM and GIS: The Development of CityGML GeoBIM Extension, <i>Advances in 3D Geo-Information Science</i>, Berlin, Springer. pp211-226. 14. CIC (2015) <i>Hong Kong BIM Standards Building Information Model - Project Execution Plan (BIM PEP)</i> Construction Industry Council, HKSAR 15. Accountants, C. P., & Bureau, B. B. (2011). <i>Project Management Accounting: Budgeting, Tracking, and Reporting Costs and Profitability</i>, by Kevin R. Callahan, Gary S. Stetz and Lynne M. Brooks Copyright© 2011 John Wiley & Sons, Inc. balance, 64(65), 143. 16. Kerzner, H. (2025). <i>Project management: a systems approach to planning, scheduling, and controlling</i>. John Wiley & Sons. 17. Meredith, J. R., Shafer, S. M., & Mantel Jr, S. J. (2017). <i>Project management: a strategic managerial approach</i>. John Wiley & Sons. 18. Ogaja, C. (2016). <i>Geomatics engineering: a practical guide to project design</i>. CRC Press.
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